

Course on Probabilistic Design of Coastal Structures

Statistical Modelling of Environmental Conditions

03. DWT ALR

Jared Ortiz-Angulo Cantos

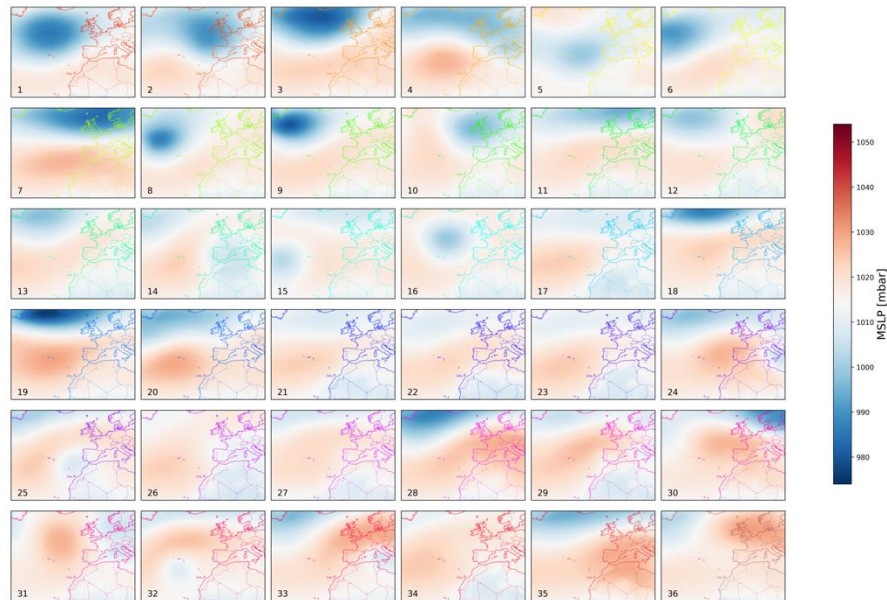
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ProCoast 2025
October 20th-23th, 2025



Introduction

DWT



DWT



N Simulations x n years

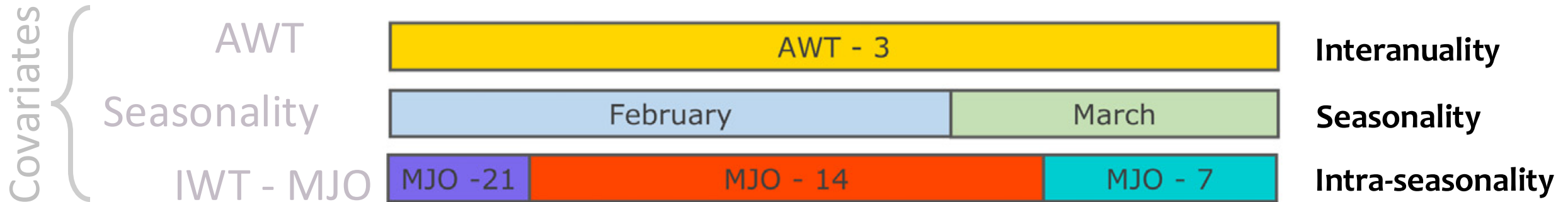
Framework

CRONOLOGY MODEL

PREDICTOR (DWT)

Anderson et al 2019
Cagigal et al 2020

Autoregressive logistic regression model



Synthetic series

$$\text{DWT} \rightarrow f(\text{AWT}, \text{Seasonality}, \text{MJO})$$

DWT

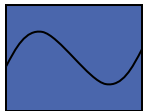


Framework

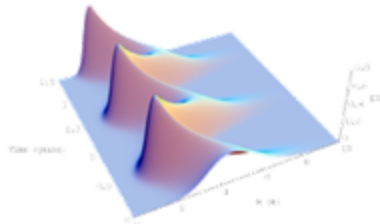
Autoregressive logistic regression model

Cagigal et al, 2020

$$\begin{aligned}
 \text{Prob}(Y_t = i | Y_{t-1}, \dots, Y_{t-e}, X_t) = \\
 = \frac{\exp(\beta_{0,i} + \beta_{1,i} \cos(\omega t) + \beta_{2,i} \sin(\omega t) + \sum_{j=1}^{n_{awt}} \beta_{j,i}^{awt} PC_j^{AWT}(t) + \sum_{j=1}^{n_{mjo}} \beta_{j,i}^{mjo} PC_j^{MJO}(t) + \sum_{j=1}^e Y_{t-j} \gamma_{j,i})}{\sum_{k=1}^{n_{DWT}} \exp(\beta_{0,k} + \beta_{1,k} \cos(\omega t) + \beta_{2,k} \sin(\omega t) + \sum_{j=1}^{n_{pc}} \beta_{j,k}^{PC} PC_j^{AWT}(t) + \sum_{j=1}^{n_{mjo}} \beta_{j,k}^{mjo} PC_j^{MJO}(t) + \sum_{j=1}^e Y_{t-j} \gamma_{j,k})}
 \end{aligned}$$



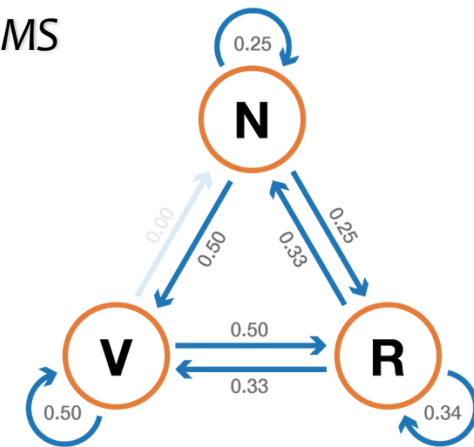
SEASONALITY



$$\theta_i(t) = \beta_0 + \sum_{i=1}^P [\beta_{2i-1} \cos(ikt) + \beta_{2i} \sin(ikt)]$$

MARKOV TERMS

3rd order

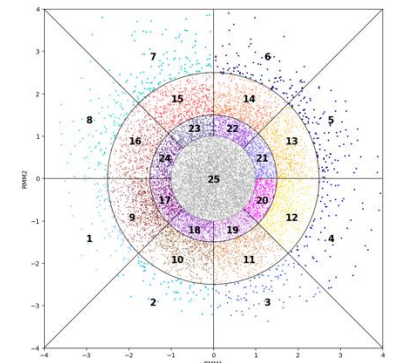
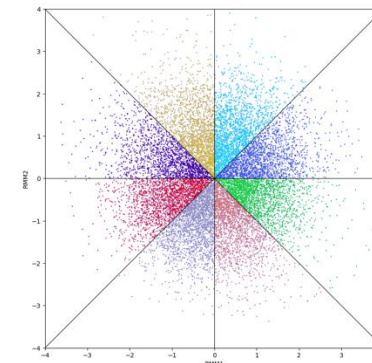
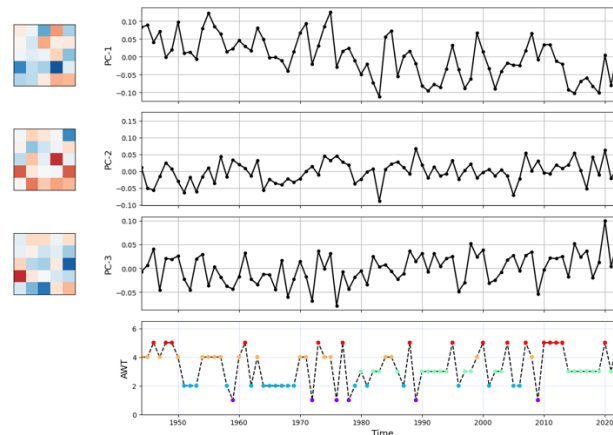


Framework

Autoregressive logistic regression model

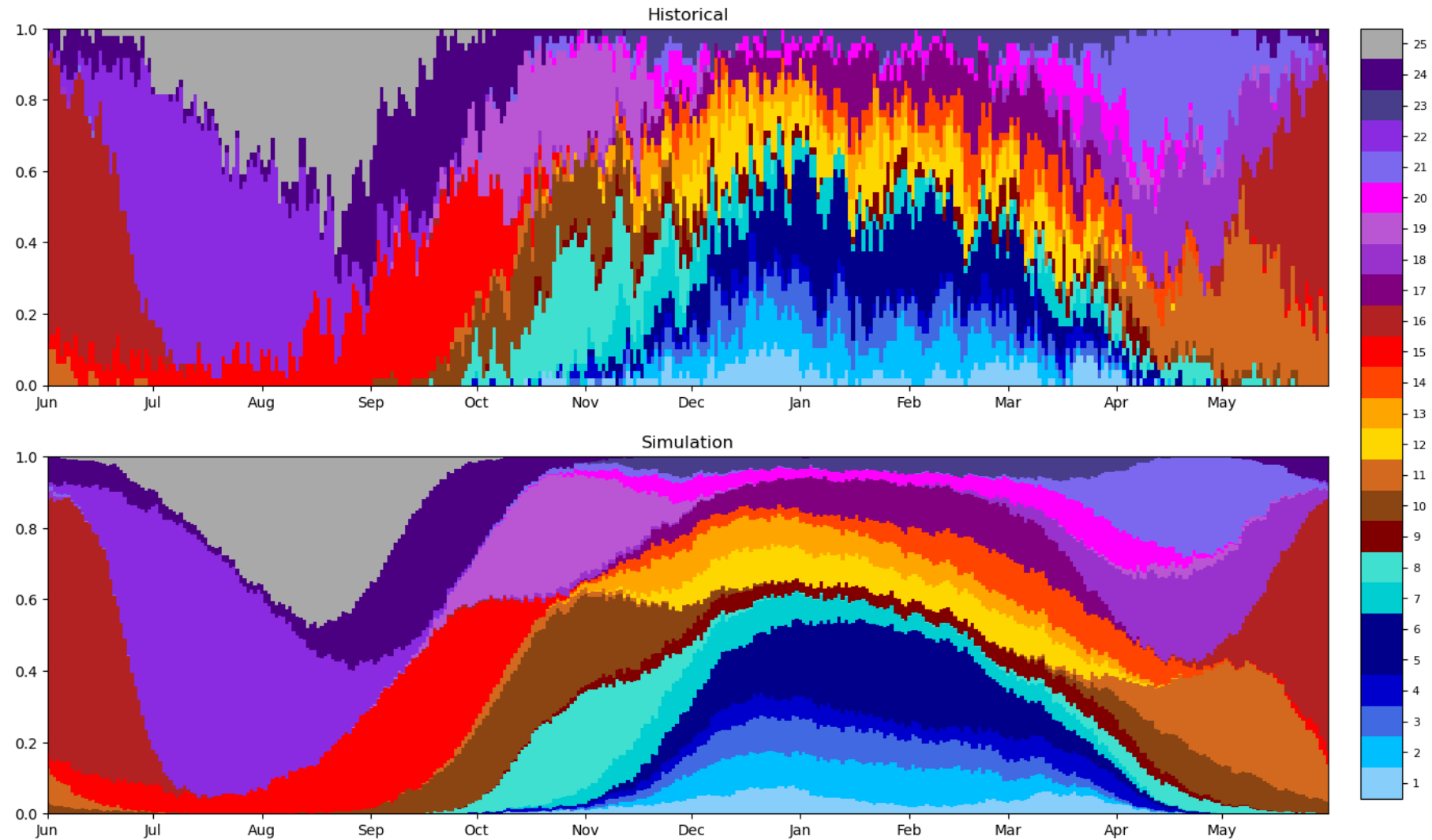
Cagigal et al, 2020

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 \end{aligned}$$



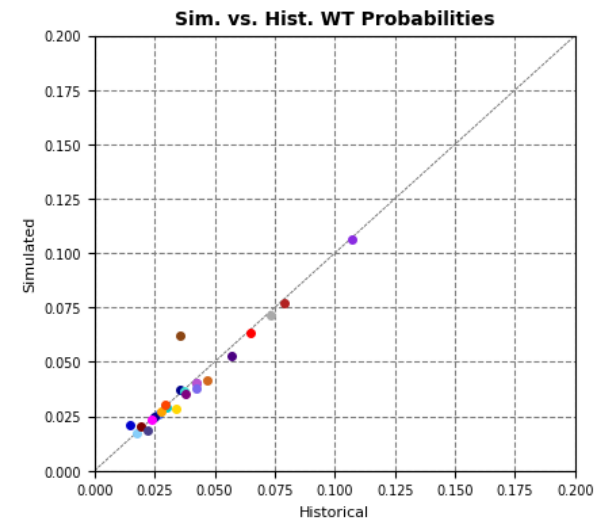
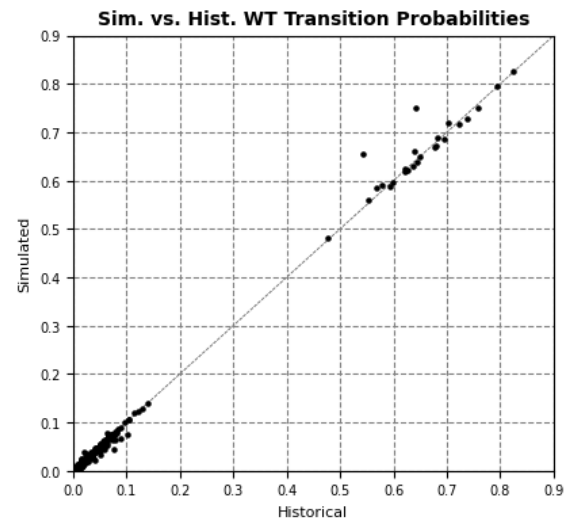
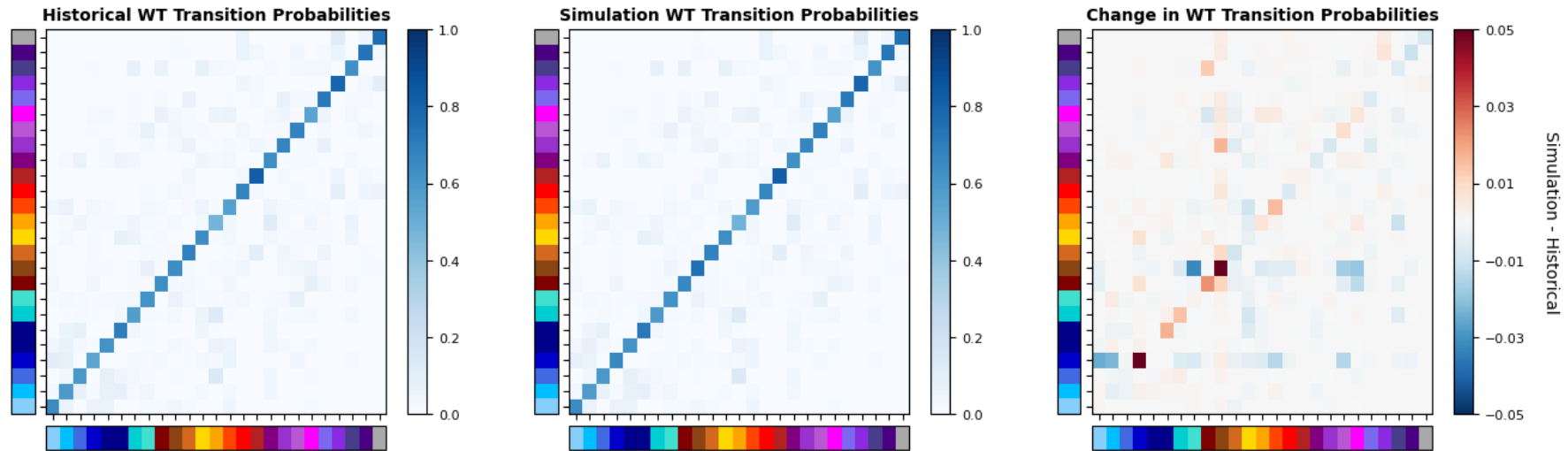
Validation

Perpetual Year



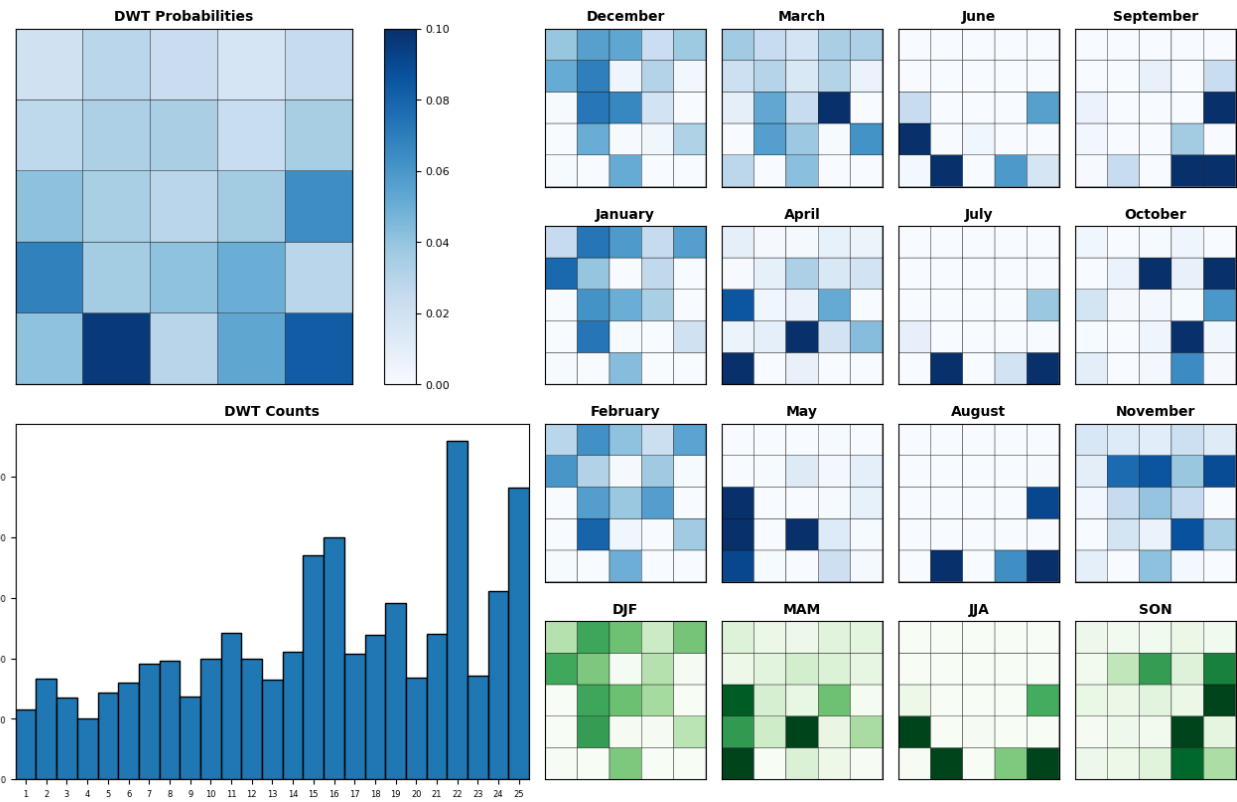
Validation

Cluster Probabilities Transitions: All Simulations

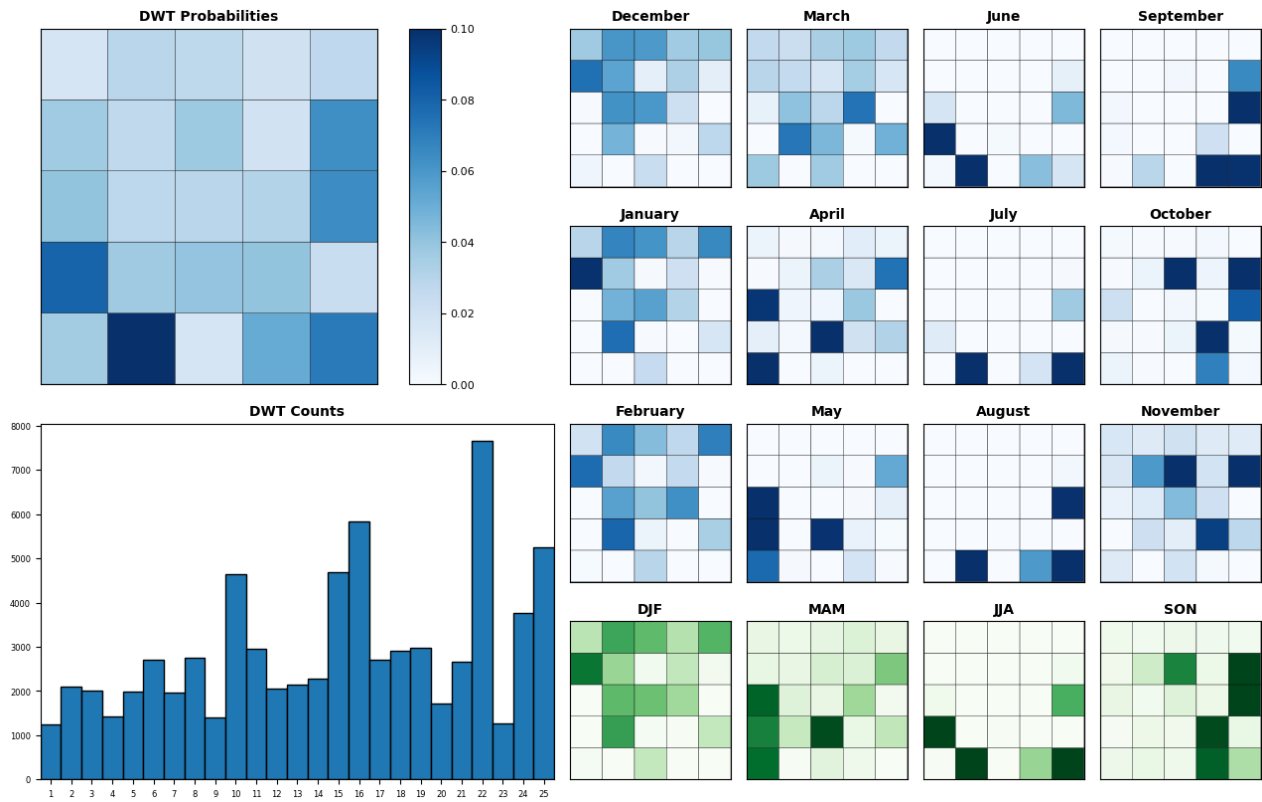


Validation

Historical



Simulated DWTs Probabilities



Thank you

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<https://geocean.sci.unican.es/hyweb/>



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